

Personalized Networking Assistant

Project Description

The Personalized Networking Assistant is an AI-powered web application designed to help users build meaningful conversations during professional networking events. It analyzes event descriptions using **DistilBERT** to identify relevant themes and generates personalized conversation starters with GPT-2. The application also provides quick fact verification through the Wikipedia API to improve conversation accuracy. Built with **FastAPI** and **Streamlit**, it offers a responsive and user-friendly interface. The system stores conversation history and user feedback for future reference and improvement.

Its modular architecture ensures scalability, maintainability, and easy integration of new features. The application was tested using **Pytest** and API testing tools to ensure reliable performance and successful request handling. It provides an efficient solution for improving networking confidence and communication.

Scenarios

Scenario 1: Smart Conversation Generation

A user enters an event description and personal interests. The system analyzes the event and generates personalized conversation starters for effective networking.

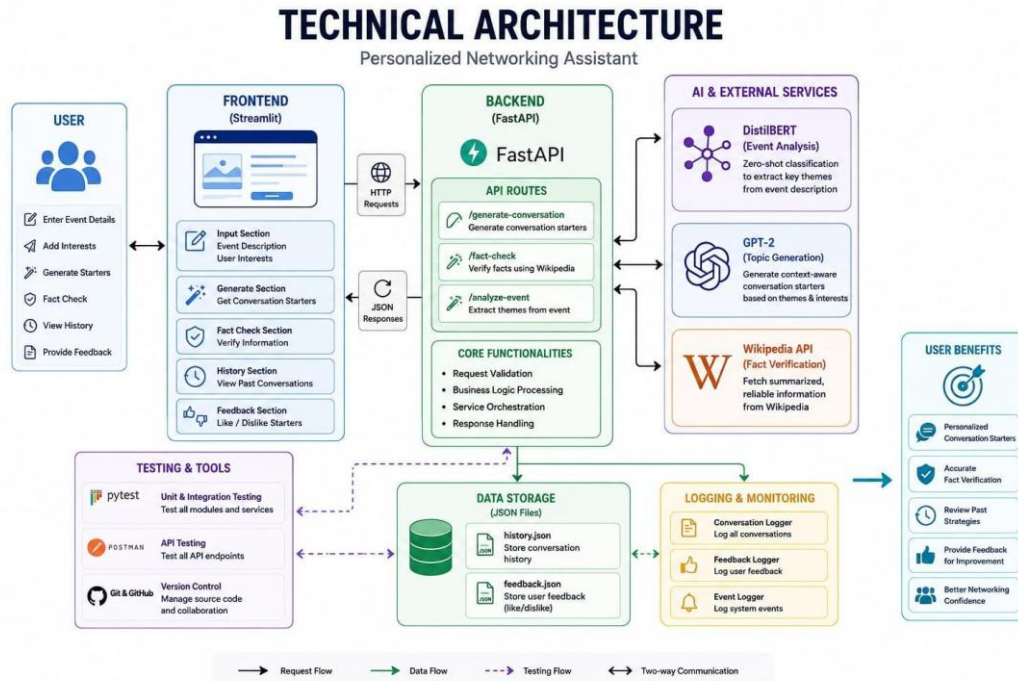
Scenario 2: Fact Verification

A user wants to verify information related to an event topic. The application retrieves a reliable summary using the Wikipedia API before the conversation.

Scenario 3: Conversation History & Feedback

A user reviews previously generated conversation starters and provides feedback. The system stores the history and feedback to support future improvements.

Technical Architecture



Description:

The Personalized Networking Assistant follows a modular three-tier architecture.

- The **Frontend Layer** consists of a Streamlit-based user interface.
- The **Backend Layer** is implemented using FastAPI and handles request processing and API routing.
- The **Service Layer** contains reusable modules for event analysis, topic generation, fact verification, history logging, and feedback management.

The architecture enables scalable, modular, and maintainable application development.

Pre-requisites

- Python Programming Knowledge
- FastAPI Framework Knowledge
- Streamlit Framework Knowledge
- Natural Language Processing Concepts
- Hugging Face Transformers Library
- REST API Concepts
- Git and GitHub Basics
- Virtual Environment Management

Project Workflow

Milestone 1: Project Planning and Environment Setup

Activity 1.1: Define Project Requirements

- > Identify networking assistant objectives..
- > Analyze user requirements and use cases.
- > Finalize project scope and features.

Activity 1.2: Design System Architecture

- > Create three-tier architecture design.
- > Define frontend, backend, and service modules.
- > Plan API communication workflow.

Activity 1.3: Configure Development Environment

- > Install Python and required libraries.
- > Set up virtual environment.
- > Configure FastAPI, Streamlit, and Transformers.

Activity 1.4: Create Project Structure

- > Create folders for app, services, models, routers, and frontend.
- > Initialize project configuration files.
- > Set up testing and documentation folders.

Milestone 2: AI Model Integration and Backend Development

Activity 2.1: Develop Event Analyzer

- > Integrate DistilBERT model.
- > Extract event details.
- > Validate user input.

Activity 2.2: Build Conversation Generator

- > Integrate GPT-2 model.
- > Generate personalized conversation starters.
- > Optimize response quality.

Activity 2.3: Implement Fact Verification

- > Integrate Wikipedia API.
- > Verify networking topics.
- > Return factual summaries.

Activity 2.4: Develop Backend APIs

- > Create FastAPI endpoints.
- > Connect backend services

Milestone 3: Frontend Development and Data Management

Activity 3.1: Develop Streamlit User Interface

- > Design interactive user interface.
- > Create input forms for event details.
- > Display conversation suggestions.

Activity 3.2: Implement History and Feedback Management

- > Store conversation history.
 - > Record user feedback.
 - > Display previous interactions.
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Milestone 4: Testing and Performance Validation

Activity 4.1: Perform Testing

- > Execute unit tests using Pytest.
- > Validate API endpoints.
- > Verify application functionality.Type your text

Activity 4.2: Analyze Performance Results

- > Measure API response time.
- > Monitor CPU and memory usage.
- > Record test results and observations.Type your text

Milestone 5: Deployment and Project Completion

Activity 5.1: Deploy the Application

- > Run FastAPI backend locally.
- > Launch Streamlit application.
- > Verify end-to-end functionality.

Activity 5.2: Final Documentation and Project Submission

- > Prepare project documentation.
 - > Capture output screenshots.
 - > Submit the final project deliverables.
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Milestone 6: Final Validation and Verification

Activity 6.1: Validate Complete System

- > Verify all application modules.
 - > Check end-to-end workflow.
 - > Confirm expected outputs. Type your text
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Activity 6.2: Review Project Deliverables

- > Verify documentation accuracy.
 - > Organize project files.
 - > Complete final project submission.
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Exploring the application pages

Home Page

The homepage allows users to:

- Enter event descriptions.
 - Specify interests.
 - Generate AI-powered conversation starters.
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Topics Section

Displays AI-detected topics extracted from the event description.

Example:

Artificial Intelligence

Machine Learning

Cloud Computing

Startups

Feedback Page

- Users can enter Session ID
 - Provide a rating(1-5)
 - Submit feedback store to improve future suggestions. ,
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Fact Checker Page

Users can verify information related to any discussion topic using Wikipedia.

History Section

Displays previously generated networking sessions.

Conclusion

Personalized Networking Assistant successfully demonstrates how Artificial Intelligence and Natural Language Processing can improve professional networking experiences. The system assists users in preparing for networking events by generating personalized conversation starters, detecting relevant topics, verifying information, and maintaining interaction history.

The modular architecture built using **FastAPI** and **Streamlit** ensures maintainability and scalability. Future enhancements include cloud deployment, user authentication, database integration, multilingual support, and integration with advanced Large Language Models such as Gemini or GPT.

This project highlights the practical application of AI in enhancing communication, professional growth, and networking effectiveness.